MILITARY SPECIFICATION

SUBASSEMBLIES, PLUG-IN, FOR TEST SET, TRANSISTOR: AN/USM - 136

1. SCOPE

1.1 This specification covers the plug-in subassemblies used in conjunction with the Transistor Set AN/USM-136 which is one type of test set used to test transistors, semiconductor diodes and other semiconductor devices.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-T-55202	Test Set, Transistor AN/USM-136
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

A8625349	Assembly, Test Set, Transistor; AN/USM-136
F8625324	Readout Converter
F8625337	h_{FE} (pulse) Base Drive Circuit
F8625596	Scaling Amplifier
F8625340	Comparison Converter
F8625341	Readout Counter
F8625342	Light Driver
D8625285	h_{fe} Base Drive Circuit
C8625532	Cross Reference, Electrical Schematics

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(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article testing.- Requirements for the submission of first article samples shall be as specified in the contract.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Fabrication.- Each subassembly shall be manufactured in accordance with its respective drawing listed in 2.1 and all drawings and specifications pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance, and permanence to insure inherent quality and performance capabilities.

3.4 General specification.- The requirements of MIL-F-13926 listed below shall be met wherever applicable.

a. Dimensions and tolerances
b. Inorganic protective surface finishes
c. Part identification and marking

3.5 Environmental.

3.5.1 Operating temperatures.- Each subassembly subjected to operating temperature requirements shall conform to the performance requirements of this specification, when thermally stabilized at ambient temperatures of -20°C (-4°F) to +50°C (+122°F).

3.5.2 Storage temperatures.- Each subassembly shall meet its respective performance requirements of this specification upon being thermally stabilized at ambient temperatures of -62°C (-80°F) and +71°C (+160°F).

3.5.3 Operating altitude.- Each subassembly subjected to operating temperature requirements shall conform to the performance requirements of this specification, when thermally stabilized at a temperature of -20°C (-4°F) and a pressure of 20.58" Hg. (10,000 ft. altitude).

3.5.4 Non-operating altitude.- Each subassembly shall meet its respective performance requirements of this specification after being subjective to a pressure of 3.44" Hg. (50,000 ft. altitude) and a temperature of -54°C (-65°F). The performance test shall be conducted at the operating altitude parameters of 3.5.3.

3.5.5 Temperature shock.- Each subassembly shall be capable of meeting the respective performance requirements of this specification upon return to room temperature of +25°C (77°F) after being subjected to a temperature shock test, the limits of which are -40°C (-40°F) and +85°C (+185°F).

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3.5.6 Shock.- Each subassembly shall be capable of withstanding 18 shocks of 15 G's acceleration for a duration of 11 plus or minus 1 milliseconds. The subassembly shall not suffer damage or subsequently fail to provide the performance specified in this specification.

3.5.7 Vibration.- Each subassembly shall be capable of meeting its respective performance requirements of this specification after being subjected to a simple harmonic motion at .03 inch amplitude (.06 inch total excursion) and frequency varying between 10 and 55 cps each minute for a total of 10 minutes duration in each of three mutually perpendicular planes.

3.5.8 Humidity.- Each subassembly shall meet its respective performance requirements of this specification after being subjected to ten cycles of the humidity test described in 4.6.1.7 of this specification.

3.6 Performance requirements.

3.6.1 Subassemblies.- The following subassemblies shall meet the requirements of their respective individual electrical performance requirements of this specification:

Plug-in Subassemblies	Frankford Arsenal Dwg. Number
(1) Readout converter (A400)	F8625324
(2) Comparison converter (A1000)	F8625340
(3) Readout counter (A1100)	F8625341
(4) Light driver (A1200)	F8625342
(5) Scaling amplifier (A900)	F8625596
(6) h_{fe} Base Drive circuit (A500)	D8625285
(7) h_{FE} (pulse) Base Drive circuit (A600)	F8625337

3.6.2 Circuitry.- The individual plug-in subassembly circuitry shall be in accordance with electrical schematics specified on Drawing C8625532, Cross Reference.

3.6.3 Warm-up.- Each subassembly shall be fully operable within 2 minutes after all the required power is applied to it.

3.6.4 Readout converter.- The Readout Converter (A400) plug-in subassembly, when used in conjunction with either the test circuit of Figure 1 or when installed in the Transistor Test Set AN/USM-136 Drawing F8625349 shall accept a positive DC input voltage over the range of 0.1 volts to 10 volts and upon the application of a negative trigger pulse of 1.5 volts that is no wider than 15 microseconds at the half voltage point shall generate two outputs. One output consists of a single positive going pulse whose width is proportional to the input DC voltage by the ratio of 1 volt equals 0.2 milliseconds. The accuracy of the voltage-to-pulse-width coversion shall be plus or minus 0.01 millisecond or plus or minus 2 percent of the pulse width, whichever is greater. The amplitude of the pulse shall be no less than 5 volts. The second output of the Readout Con-

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verter consists of a train of 50 Kc pulses of an amplitude no less than 8 volts. The number of 50 Kc pulse is proportional to the input DC voltage by the ratio of 1 volt equals 10 pulses. The accuracy of the voltage-to-pulse-quantity conversion shall be plus or minus 2 pulses for any input voltage over 1.0 volts and plus or minus 1 pulse for any input voltage 1.0 volt or below when tested at room temperature. Add plus or minus 1 pulse to above accuracy values for tests conducted at high or low temperature. A reset pulse shall be of negative polarity and an amplitude of 3 volts with a pulse width 25 microseconds wide at the half voltage point.

3.6.5 Comparison converter.- The Comparison Converter (A1000) plug-in subassembly, when used in conjunction with either the test circuit of Figure 2 or when installed in the Transistor Test Set AN/USM-136 shall accept a positive DC input voltage over the range of 0.1 volt to 10. volts and upon application of a negative trigger pulse of 1.5 volts that is no wider than 15 microseconds at the half voltage point, shall generate for its internal use a single pulse whose width is proportional to the input voltage by the ratio of 1 volt equals 0.2 milliseconds. The accuracy of this voltage-to-pulse-width conversion shall be plus or minus 0.01 millisecond or plus or minus 2 percent of the pulse width, whichever is greater.

3.6.5.1 Simultaneously with the application of the trigger, the single pulse from the Readout Converter is fed into the Comparison Converter and in conjunction with an external HIGH-LIMIT, LOW-LIMIT relay, is compared with the width of the internally generated pulse for a GO, NO-GO decision.

3.6.5.2 A 20 microsecond difference in the two pulse widths shall be recognized by this comparison circuit to actuate the NO-GO decision. A separate direct NO-GO input pulse of at least 12 volts negative shall be capable of actuating the NO-GO decision. The output of this comparison converter subassembly will close an external 28 volt 700 ohm relay to provide the GO, NO-GO indication. A reset pulse shall be of negative polarity and an amplitude of 3 volts with a pulse width 25 microseconds wide at the half voltage point.

3.6.6 Readout counter.- The Readout Counter (A1100) plug-in subassembly, when used in conjunction with either the test circuit of Figure 3 or when installed in the Transistor Test Set AN/USM-136 shall accept as an input a train of pulses from as high as 50 Kc to as low as 5 Kc and of an amplitude of as low as 7 volts positive going and shall produce as an output, a conversion to modified binary (decimal) output signal of sufficient amplitude to drive the NOR logic circuits of the Light Driver F8625342. The output amplitude shall be greater than 7 volts. A reset pulse of at least 3 volts negative that is wider than 25 microseconds at the half voltage point shall reset all counters to zero.

3.6.7 Light driver.- The Light Driver (A1200) plug-in subassembly, when used in conjunction with either the test circuit of Figure 3 or when installed in the Transistor Test Set AN/USM-136 shall accept as an input the modified binary signals developed by the Readout Counter or as simulated

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by the four toggle switches in Figure 7 and operated in accordance with chart and convert them into decimal numbers and switch on the lamp drivers to light the external 28 volt lamp in the projection readout unit directly associated with the proper sequence of decimal decade count.

3.6.8 Scaling amplifier.- The Scaling Amplifier (A900) plug-in subassembly, when used in conjunction with either the test circuit of Figure 4 or when installed in the Transistor Test Set AN/USM-136 shall accept an AC input signal of either 60 cps, sinewave or square wave, 300 microsecond pulses or 1 Kc sinewave of an amplitude from 0.0001 volt to 10 volts. The output DC voltage can be kept within the range of from 0.1 volt to 10 volts in conjunction with the external programming relays to enable the input signal to be subjected to a gain of X1, X10, X100, and X1,000. The gain accuracy or ratio of AC input to DC output for each range shall be plus or minus 2 percent full scale. The input impedance shall be no less than 1 megohm.

3.6.9 h_{fe} Base drive circuit.- The h_{fe} Base Drive Circuit (A500) plug-in subassembly, when used in conjunction with the test circuit of Figure 5, or when installed in the Transistor Test Set AN/USM-136 shall generate a 1,000 plus or minus 20 cycle per second, sinewave current signal. The output magnitude of this circuit shall be programmable in three current ranges: 0.2 microamperes peak-to-peak, 2.0 microamperes peak-to-peak, or 20 microamperes peak-to-peak. Each range has an accuracy of plus or minus 2 percent.

3.6.10 h_{FE} (pulse) Base drive circuit.- The h_{FE} (pulse) Base Drive Circuit (A600) plug-in subassembly, when used in conjunction with the test circuit of Figure 6, or when installed in the Transistor Test Set AN/USM-136 shall generate a pulse train of constant amplitude pulses 350 plus or minus 150 microseconds wide at a frequency of 35 plus or minus 10 cycles per second. By the use of two external relays twelve (12) different programmed values of pulse current output shall be available for base drive:

1.0 amperes ±2%	100 milliamperes ±2%	10 milliamperes ± 2%
500 milliamperes ±3%	50 milliamperes ±2%	5 milliamperes ±3%
250 milliamperes ±4%	25 milliamperes ±2%	2.5 milliamperes ±4%
125 milliamperes ±5%	12.5 milliamperes ±2%	1.25 milliamperes ±5%

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article testing.- When the contract specified testing of preproduction or initial production samples, the following paragraphs shall apply as applicable.

4.2.1 Preproduction test.- Preproduction test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3, 4.4 and 4.6 as well as any additional tests that may be specified in the contract.

4.2.2 Initial production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4 and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

4.2.3 First article disapproval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as required by the contract clauses applicable to first article testing.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, and the sampling plans specified in Table I herein. Examinations and tests for Packaging, Packing, and Marking shall be in accordance with MIL-P-14232. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL 1.0% Defective		

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TABLE I - CLASSIFICATION OF DEFECTS (Cont'd)

		REQUIREMENT	TEST PROCEDURE
101.	Warm-up	3.6.3	4.6.2.1
102.	Readout converter	3.6.4	4.6.3
103.	Comparison converter	3.6.5	4.6.4
104.	Readout counter	3.6.6	4.6.5
105.	Light driver	3.6.7	4.6.6
106.	Scaling amplifier	3.6.8	4.6.7
107.	h_{fe} (AC Beta) Base drive circuit	3.6.9	4.6.8
108.	h_{FE} (DC Beta pulsed) Base drive circuit	3.6.10	4.6.9

MINOR: NONE

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.- Three units shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
401.	Fabrication	3.3	Applicable drawing - visual
402.	General specification	3.4	MIL-F-13926 - visual

4.4.1 Environmental.- Three units shall be selected at random by a Government representative as a standard sample from each 50 produced or from each month's production, whichever occurs first. The standard samples shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table III.

TABLE III

	CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
301.	Extreme temperatures	3.5.2	4.6.1.1, 4.6.1.2
302.	Altitude	3.5.3, 3.5.4	4.6.1.3
303.	Temperature shock	3.5.5	4.6.1.4
304.	Shock	3.5.6	4.6.1.5
305.	Vibration	3.5.7	4.6.1.6
306.	Humidity	3.5.8	4.6.1.7

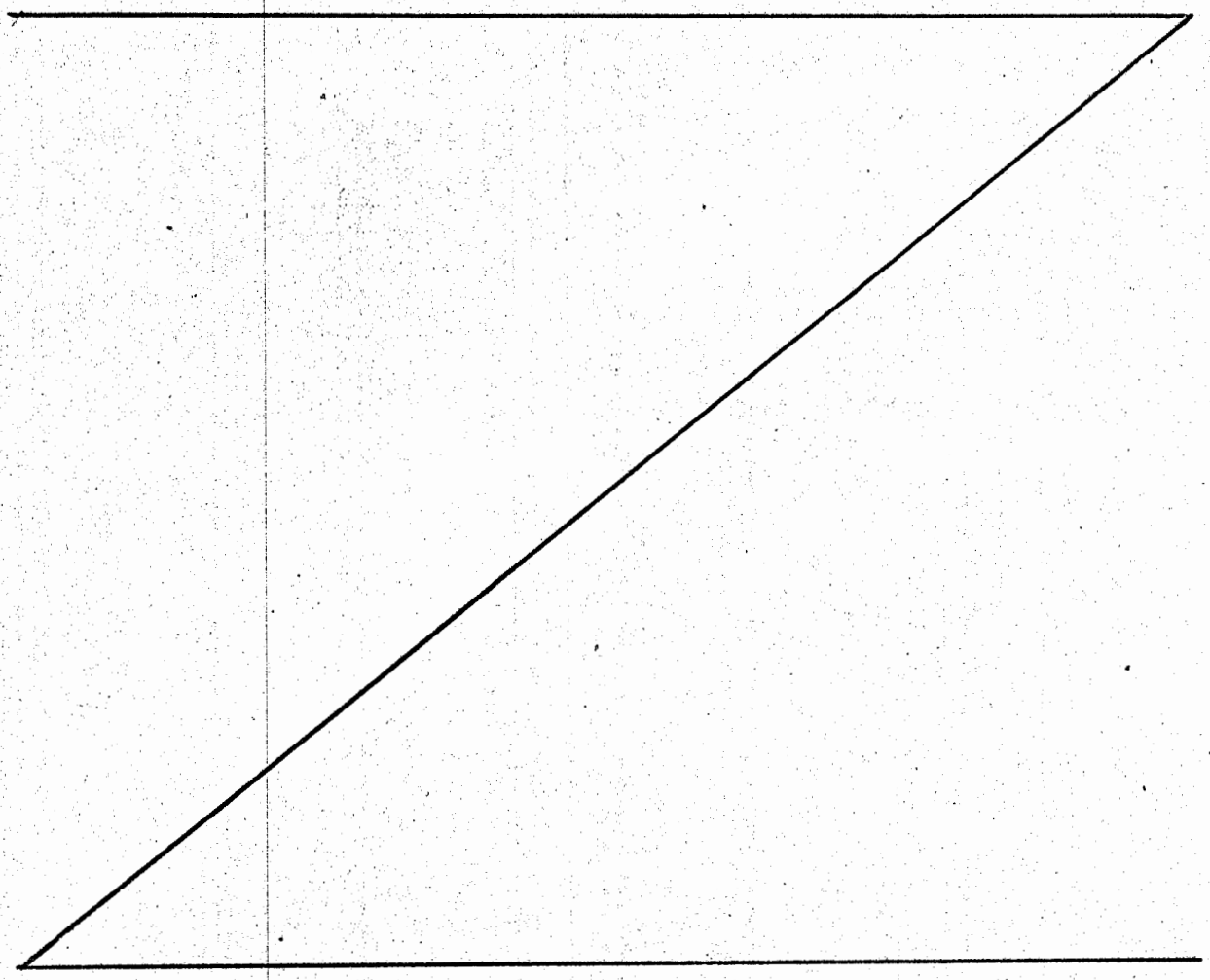
4.4.2 Failure of sample.- Should any one item of a standard sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

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4.5.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.1.1 Test equipment.- In conjunction with 4.5.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.6:



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Item	Description	Accuracy Requirements
1.	Power Supply	Negative 28 volts plus or minus 10 percent, current rating greater than or equal to 500 milliamperes, peak-to-peak ripple less than 1 volt, output impedance less than 10 ohms from DC to 20 kilocycles.
2.	Power Supply	Negative 24 volts plus or minus 1 percent, current rating greater than or equal to 100 milliamperes, peak-to-peak ripple less than 0.05 volt, output impedance less than 2 ohms from DC to 500 kilocycles.
3.	Power Supply	Negative 12 volts plus or minus 1 percent, current rating greater than or equal to 50 milliamperes, peak-to-peak ripple less than 2 ohms from DC to 500 kilocycles.
4.	Power Supply	Positive 8 volts plus or minus 5 percent, current rating greater than or equal to 100 milliamperes, peak-to-peak ripple less than 0.05 volt, output impedance less than 2 ohms from DC to 500 kilocycles.
5.	Power Supply	Positive 12 volts plus or minus 1 percent, current rating greater than or equal to 50 milliamperes, peak-to-peak ripple less than 0.05 volt, output impedance less than 2 ohms from DC to 500 kilocycles.
6.	Power Supply	Positive 24 volts plus or minus 1 percent, current rating greater than or equal to 100 milliamperes, peak-to-peak ripple less than 0.05 volt, output impedance less than 2 ohms from DC to 500 kilocycles.
7.	Pulse Generator	Reset pulse generator of negative 3 volts having a rise time of no greater than 1 microsecond and a pulse width 25 microseconds wide at the half voltage point.
8.	Pulse Generator	Trigger pulse generator of negative 1.5 volts having a rise time of no greater than 1 microsecond and a pulse width 15 microseconds wide at the half voltage point.
9.	Square Wave Generator	Input square wave generator of 7 volt amplitude having a positive going rise time of no greater than 1 microsecond and a variable frequency range of from 5 kilocycles to 50 kilocycles.

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Item	Description	Accuracy Requirements
10.	Signal Generator	DC input signal continuously variable from positive 0.1 volts to positive 10 volts having a 1.2 microfarad capacitor across its output terminals.
11.	Signal Generator	AC input signal continuously variable from 0.0001 volt peak-to-peak to 10 volts peak-to-peak, 60 cycles per second square wave having rise and fall times not to exceed 2 microseconds.
12.	Oscilloscope	Passband; DC to 30 megacycles calibrated deflection factor: 0.005 to 20 volts per centimeter Fastest sweep (magnified): 20 nanosecond per centimeter Slowest sweep (calibrated): 5 second per centimeter, ±3% accuracy Calibrated sweep magnifier: 5X Cathode ray tube: accelerating voltage 10 kilovolts Vertical scale: 4 centimeter
13.	Test Resistors	1 megohm ±1% 1/2 watt 100 K ohm ±1% 1/2 watt 10 K ohm ±1% 1/2 watt 1 K ohm ±1% 1/2 watt 100 ohm ±1% 1/2 watt 10 ohm ±1% 1/2 watt 1 ohm ±1% 1/2 watt

4.6 Test methods and procedures.

4.6.1 Environmental.

4.6.1.1 Low temperature.

4.6.1.1.1 Low temperature Procedure I.- The plug-in subassembly shall be placed within the test chamber and the chamber cooled to and maintained at a temperature of -20°C (-4°F) until stabilization of the subassembly temperature is reached. While at this temperature, the subassembly shall be energized tested to determine conformance to its requirements of this specification.

4.6.1.1.2 Low temperature Procedure II.- Following the test in 4.6.1.1.1, the temperature within the chamber shall be reduced to and maintained at a temperature of -62°C (-80°F) for 72 hours with the subassembly de-energized. The temperature of the chamber shall then be raised slowly to the laboratory ambient temperature and the equipment shall meet its performance requirements of this specification.

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4.6.1.2 High temperature.- The plug-in subassembly shall be placed within the test chamber and the chamber temperature raised to +71°C (+160°F). The relative humidity within the chamber shall not be more than 15%. The subassembly shall be maintained at this temperature for a period of 48 hours. At the conclusion of the exposure period, the temperature within the chamber shall be lowered to +50°C (+122°F). After equipment stabilization at +50°C (+122°F), the subassembly shall be energized and operated at this temperature and shall meet its performance requirements of this specification. The chamber temperature shall then be lowered to the laboratory ambient temperature and a visual inspection performed to determine if any deterioration has occurred.

4.6.1.3 Altitude.- The plug-in subassembly shall be placed in the test chamber and the internal chamber temperature reduced to -54°C (-65°F). The internal pressure of the chamber shall be reduced to 3.44 In. Hg. (50,000 ft. altitude). The subassembly shall be maintained under these conditions for one hour. At the conclusion of this period, the internal chamber temperature shall be raised to -20°C (-4°F) and the internal pressure of the chamber shall be raised to 20.58 In. Hg. (10,000 ft. altitude). After the subassembly has stabilized at the new temperature and pressure, the subassembly shall be energized and shall meet its performance requirements of this specification. The chamber shall then be returned to laboratory ambient and pressure and the subassembly shall be visually inspected for damage.

4.6.1.4 Temperature shock.- The plug-in subassembly shall first be placed within a test chamber wherein a temperature of +85°F, is maintained. The subassembly shall be exposed to this temperature for a period of 4 hours, at the conclusion of which and within 5 minutes, the subassembly shall be transferred to a chamber having an internal temperature of -40°C (-40°F). The subassembly shall be subjected to this temperature for a period of 4 hours. This constitutes one cycle. The number of complete cycles shall be three. The duration of exposure at each extreme temperature shall not be less than that specified and may be extended to overnight exposure to prevent interruption of the transfer sequence. At the conclusion of the third cycle, the subassembly shall be removed from the test chamber, returned to standard temperature, and within a period of 1 hour shall be visually inspected for damage and shall be energized and operated and shall meet its performance requirements of this specification.

4.6.1.5 Shock.- The plug-in subassembly, non-operating, mounted in a test fixture to simulate the way in which it is normally mounted in service shall be subjected to the shock conditions specified herein. The subassembly shall be subjected to 18 impact shocks of 15g acceleration, each shock impulse having a time duration of 11 plus or minus 1 milliseconds. The maximum acceleration shall be reached in approximately 5-1/2 milliseconds. The shock shall be applied in the following directions:

a. Vertically, 3 shocks in each direction
b. Parallel to the major horizontal axis, 3 shocks in each direction
c. Parallel to the minor horizontal axis, 3 shocks in each direction

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The subassembly shall be inspected to assure that it did not suffer damage and that it will provide the performance specified for it.

4.6.1.6 Vibration.- This test shall be conducted in accordance with Procedure I of MIL-F-13926 except that the frequency, amplitude and cycle time specified in 3.5.7 shall be applied.

4.6.1.7 Humidity.- The plug-in subassembly shall be placed in the test chamber and during the first 2 hour period, the temperature in the test chamber shall be raised from 28°C (82°F) to 71°C (160°F). The temperature of 71°C (160°F) and a relative humidity of 95 percent shall be maintained in the test chamber during the next 6 hour period. During the following 16 hour period, the temperature in the chamber shall be reduced at a uniform rate to 28°C (92°F) which constitutes one cycle. The cycle shall be repeated 10 times for a total test time of 240 hours. At the conclusion of the 240 hour period, the subassembly shall be returned to laboratory environment and shall meet its requirements of this specification. Distilled or demineralized water having a pH value between 6.5 and 7.5 at 25°C (77°F) shall be used to obtain the desired humidity.

4.6.2 Examination.- Each plug-in subassembly shall be energized and subjected to an operating test to insure qualitatively the proper functioning of the circuits and conformance with the requirements of 3.6 of this specification.

4.6.2.1 Test conditions.- Unless otherwise specified, all tests shall be performed under the following conditions:

(a) Ambient conditions; the tests shall be performed under factory or laboratory ambient conditions.

(b) Input power; the input power shall be as shown in the appropriate test circuit figure or 115 volts plus or minus 2 percent 60 cycles per second plus or minus 2 percent, if tested in the Transistor Test Set AN/USM-136.

(c) Warm-up; the subassemblies shall be turned ON and permitted to warm up for 2 minutes before any test is conducted.

4.6.3 Readout converter.- Testing the readout converter to meet the requirements of 3.6.4 shall be done in the following manner:

(a) Insert the plug-in subassembly Readout Converter, (A400) into the test circuit of Figure 1 and connect the power supplies and other items of test equipment as indicated (or test while installed in the Transistor Test Set AN/USM-136). Turn on the power and allow a 2 minute warm-up period. If the following tests are conducted at other than normal room temperature, add plus or minus 1 pulse to all the acceptance limits in (d), (e), and (h).

(b) A DC positive voltage of 0.2 volts shall be applied on input Pin 23 of J401. Apply a negative 3.0 volt reset pulse 25 microseconds wide at the half voltage point to Pin 1.

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(c) Apply a negative 1.5 volt trigger pulse 15 microseconds wide at the half voltage point to Pin 17 of J401 and observe on an oscilloscope connected to Pin 13 that a positive going pulse of no less than 5 volts amplitude at a pulse width of 0.04 milliseconds plus or minus 0.01 milliseconds has been produced.

(d) With the same input voltage, reset pulse, and trigger pulse, as in (b) and (c) above. Observe on an oscilloscope connected to Pin 15 that a train of two (2) pulses plus or minus 1 pulse at a 50 Kc rate of an amplitude of no less than 8 volts has been produced.

(e) Change the input voltage to 1.0 volts, reset and trigger as above and observe on an oscilloscope connected to Pin 15 that a train of ten (10) pulses plus or minus 1 pulse at a 50 Kc rate of an amplitude of no less than 8 volts has been produced.

(f) With the same input voltage, reset and trigger as in (e) above, observe on an oscilloscope connected to Pin 13 that a positive going pulse of no less than 5 volts amplitude at a pulse width of 0.2 milliseconds plus or minus 0.01 milliseconds has been produced.

(g) Change the input voltage to 9.8 volts, reset and trigger as in (f) above and observe on an oscilloscope connected to Pin 13 that a positive going pulse of no less than 5 volts amplitude at a pulse width of 1.96 milliseconds plus or minus 0.04 milliseconds has been produced.

(h) With the same input voltage, reset and trigger as in (g) above, observe on an oscilloscope connected to Pin 15 that a train of 98 pulses plus or minus 2 pulses at a 50 Kc rate of an amplitude of no less than 8 volts has been produced.

4.6.4 Comparison converter.- Testing the comparison converter to meet the requirements of 3.6.5 shall be done in the following manner:

(a) Insert the plug-in subassembly, Readout Converter (A1000), into the test circuit of Figure 2 and connect the power supplies and other items of test equipment as indicated (or test while installed in the Transistor Test Set AN/USM-136). Turn on the power and allow a 2 minute warm-up period. Place the test switch SW-1 on the LOW LIMIT position to simulate LOW LIMIT position of relay K1001 in the test set.

(b) A positive DC voltage of 0.2 volts shall be applied to the input Pin 23 of J1001. Apply a negative 3.0 volt reset pulse, 25 microseconds wide at the half voltage point to Pin 1.

(c) Apply a negative 1.5 volt trigger pulse 15 microseconds wide at the half voltage point to Pin 17 of J1001 and observe on an oscilloscope connected to Pin 14 that a positive going pulse of no less than 5 volts amplitude at a pulse width of 0.04 milliseconds plus or minus 0.01 millisecond has been produced and note the value. The "NO-GO" bulb should light.

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(d) Repeat steps (b) and (c) but simultaneous with the application of the trigger pulse apply a positive 5.0 volt pulse ± .2 volt to Pin 9 that is 20 microseconds longer than the pulse obtained in (c) above. The "NO-GO" light should not light.

(e) Place the test switch SW-1 on the HIGH-LIMIT position to simulate HIGH LIMIT position of relay K1001 in the test set and remove the signal to Pin 9.

(f) Change the DC positive voltage on the input Pin 23 to plus 9.8 volts, reset and trigger as above and observe on an oscilloscope connected to Pin 14 that a positive going pulse of no less than 5 volts amplitude at a pulse width of 1.96 milliseconds plus or minus 0.04 milliseconds has been produced and note the value. The "NO-GO" light should not light.

(g) Repeat step (f) but simultaneous with the application of the trigger pulse apply a positive 5.0 volt pulse ± .2 volt to Pin 9, that is 20 microseconds longer than the pulse obtained in (f) above. The "NO-GO" light shall light.

(h) Reapply the reset pulse.

(i) Apply a negative "NO-GO" signal by closing the test switch SW-2 to simulate the signal from the automatic lead selector - in-circuit (A100) module. Observe that the "NO-GO" light shall light.

4.6.5 Readout counter.- Testing the readout counter to meet the requirements of 3.6.6 shall be done in the following manner:

(a) Insert the plug-in subassembly. Readout Counter (A1100), into the test circuit of Figure 3 and connect the power supplies and other items of test equipment as indicated (or test while installed in the Transistor Test Set AN/USM-136). Turn on the power and allow a 2 minute warm-up period.

(b) Apply a continuous train of 50 Kc square wave pulses of a 7 volt amplitude to the input Pin 1 of J1101. Observe on an oscilloscope connected to Pin 20 of J1101 that the output consists of a train of 5 Kc pulses having an amplitude in excess of 7 volts.

(c) Change the input pulse frequency to 5 Kc at the same voltage and observe that the output shall consist of a train of 500 cycles per second pulses having an amplitude in excess of 7 volts.

(d) Stop the input signal so that "ZERO" light bulb will not light (turn off).

(e) Apply a reset pulse of negative 3 volts having a pulse width at least 25 microseconds wide at the half voltage point to Pin 24 and observe that the "ZERO" bulb will light (turn on).

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4.6.6 Light driver.- Testing the light driver to meet the requirements of 3.6.7 shall be done in the following manner:

(a) Insert the plug-in subassembly, Light Driver (A1200) into the test circuit of Figure 7 and connect the power supplies and other items of test equipment as indicated (or test while installed in the Transistor Test Set AN/USM-136). Turn on the power and allow a 2 minute warm-up period.

(b) Place all four switches in "ZERO" position, observe the "ZERO" light indication.

(c) Move SW-1 to the "ONE" position and observe that the "ZERO" light will go out and the "ONE" light will come on.

(d) Move SW-1 back to "ZERO" and move SW-2 to "ONE" and note the change from the "ONE" to the "TWO" light.

(e) Operate the four switches according to the chart and observe the change from TWO to THREE, etc. on through "NINE" and then back to "ZERO".

4.6.7 Scaling amplifier.- Testing the scaling amplifier to meet the requirements of 3.6.8 shall be done in the following manner:

(a) Insert the plug-in subassembly, Scaling Amplifier (A900), into the test circuit of Figure 4 and connect the power supplies and other items of test equipment as indicated (or test while installed in the Transistor Test Set AN/USM-136). Turn on the power and allow a 2 minute warm-up period.

(b) With the test switch SW-1 in the X1 position, apply an input signal of 9.8 volts 60 cycles per second square wave signal to the input capacitor C1 (which is connected to Pin 7 of J901). The output DC voltage measured by a 10 volt voltmeter having an accuracy of plus or minus 0.5 percent connected to Pin 21 of J901 shall read plus 9.8 volts plus or minus 2 percent.

(c) A 1 megohm plus or minus 5 percent 1/2 watt test resistor shall be connected in series between the source of input voltage and the input capacitor C1. Repeat test (b) above and note that the voltage read on the DC output voltmeter shall be greater than half of the value measured during test (b).

(d) Remove the 1 megohm test resistor and reconnect the input signal to the input capacitor C1. Change the input signal to 0.98 volts 60 cycles per second square wave and the test switch SW-1 to the X10 position. Observe that the DC output voltmeter reading is plus 9.8 volts plus or minus 2 percent.

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(e) Change the input signal to 0.098 volts, 60 cycles per second square wave and the test switch SW-1 to the X100 position and observe that the DC output voltmeter reading is plus 9.8 volts plus or minus 2 percent.

(f) Change the input signal to 0.0098 volts 60 cycles per second square wave and the test switch SW-1 to the X1000 position and observe that the DC output voltmeter reading is plus 9.8 volts plus or minus 2.

4.6.8 h_{fe} Base drive circuit.- Testing the h_{fe} base drive circuit to meet the requirements of 3.6.9 shall be done in the following manner:

(a) Insert the plug-in subassembly, h_{fe} Base Drive Circuit (A500), into the test circuit of Figure 5 and connect the power supplies and other items of test equipment as indicated (or test while installed in the Transistor Test Set AN/USM-136). Turn on the power and allow a 2 minute warm-up period.

(b) Place the test switch SW-1 in the 0.2 microampere peak-to-peak position. Use the 100K ohm plus or minus 1 percent 1/2 watt test resistor across Pins F and H of J503. Observe on an oscilloscope placed across the test resistor the presence of a 1 Kc plus or minus 20 cycles per second sine wave signal of an amplitude 20 millivolts plus 2 percent or minus 4 percent. Note: Consideration should be given to the loading effect of the oscilloscope input impedance on the equivalent value of the test resistor.

(c) Change the test switch SW-1 to the 2.0 microampere peak-to-peak position. Replace the 100K ohm test resistor with a 10K ohm plus or minus 1 percent 1/2 watt test resistor across the same pin location as in (b). Observe on the oscilloscope that the amplitude of the 1 Kc signal is now 20 millivolts plus or minus 3 percent.

(d) Change the test switch SW-1 to the 20 microampere peak-to-peak position. Replace the 10K ohm test resistor with a 1,000 ohm plus or minus 1 percent 1/2 watt test resistor across the same pin position as in (b). Observe on the oscilloscope that the amplitude of the 1 Kc, signal is still 20 millivolts plus or minus 3 percent.

4.6.9 h_{FE} (Pulse) Base drive circuit.- Testing the h_{FE} (pulse) base drive circuit to meet the requirements of 3.6.10 shall be done in the following manner:

(a) Insert the plug-in subassembly h_{FE} (pulse) Base Drive Dircuit (A600) into the test circuit of Figure 6 and connect the power supplies and other items of test equipment as indicated (or test while installed in the Transistor Test Set AN/USM-136). Turn on the power and allow a 2 minute warm-up period.

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(b) Place a 1 ohm plus or minus 1 percent 1/2 watt test resistor as shown in Figure 6 and move the test switch SW-1 to the position marked 1 and the test switch SW-2 to the 1 amp position. Observe that the pulse as seen on an oscilloscope whose input probe is placed across the 1 ohm test resistor is 350 plus or minus 75 microseconds wide at a frequency of 35 plus or minus 10 cycles per second and of an amplitude 1.0 volt plus or minus 3 percent peak-to-peak.

(c) Replace the test resistor with one of 10 ohms plus or minus 1 percent 1/2 watt in the same location. Change the test switch SW-2 to the 100 ma position. Observe on the oscilloscope that the pulse amplitude is still 1.0 volt plus or minus 3 percent peak-to-peak.

(d) Change the test switch SW-1 to the position marked .5 and observe on the oscilloscope that the pulse amplitude is now 0.5 volts plus or minus 3 percent peak-to-peak.

(e) Change the test switch SW-1 to the position marked 0.25 and observe on the oscilloscope that the pulse amplitude is now 0.25 volts plus or minus 3 percent peak-to-peak.

(f) Change the test switch SW-1 to the position marked .125 and observe on the oscilloscope that the pulse amplitude is now 0.125 volts plus or minus 3 percent peak-to-peak.

(g) Replace the test resistor with one of 100 ohms plus or minus 1 percent 1/2 watt in the same location. Change the test switch SW-1 to the position marked 1 and the test switch SW-2 to the 10 ma position. Observe on the oscilloscope that the pulse amplitude is now. 1.0 volt plus or minus 3 percent peak-to-peak.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Preservation, packaging and packing shall be in accordance with MIL-P-14232 and the applicable packaging data sheet. The levels of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The plug-in subassemblies are for use in the Test Set, Transistor AN/USM-136 which is an instrument used to test transistor and semiconductor devices which fall into the design limits of the instrument.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Applicable packaging data sheet.
c. Level of protection required.

Custodian:

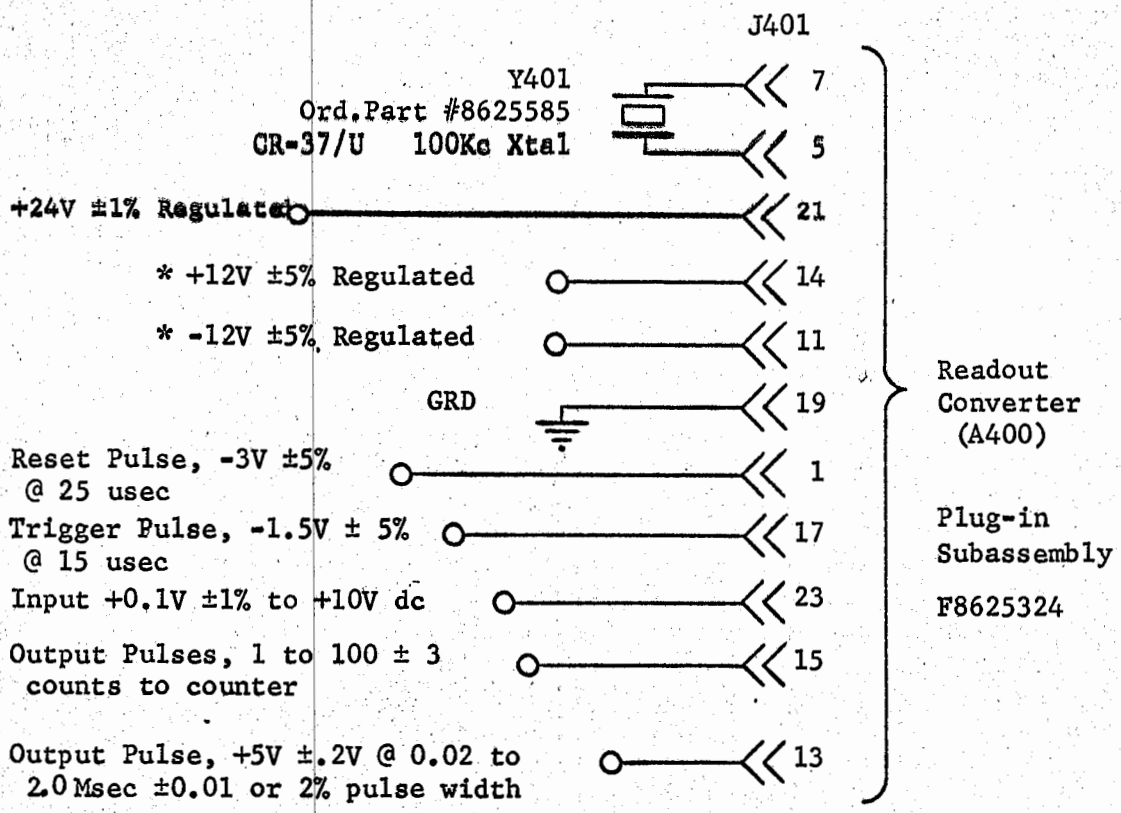
Army - MU

Preparing activity:

Army - MU

Project No. 6625-A223

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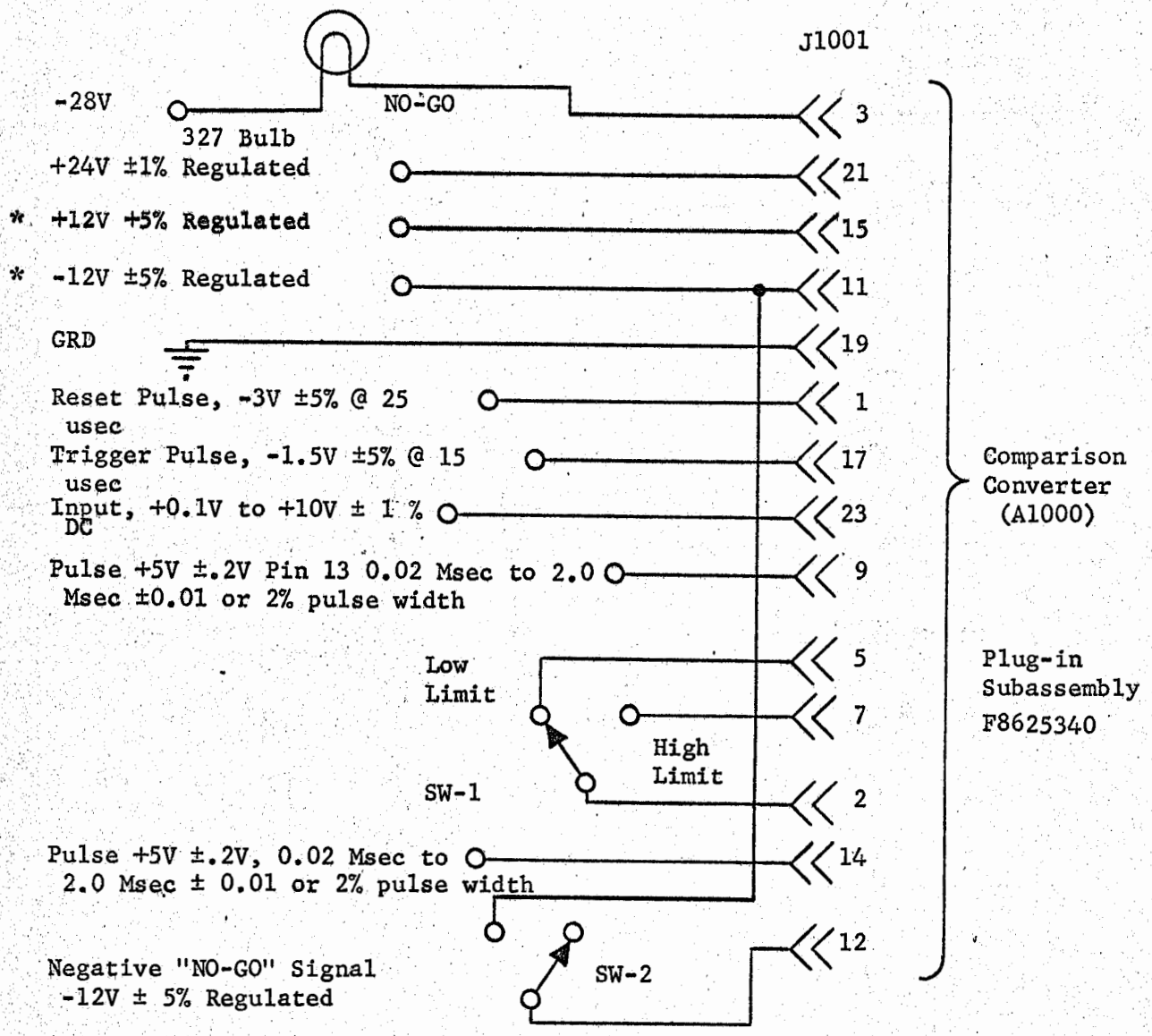


*Note: -12V must be at least 0.1 volt greater value than the +12V value -- examples; -12V and +11.9V; or -12.6V and +12.5V, or -11.6V and +11.4V.

READOUT CONVERTER TEST CIRCUIT

FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
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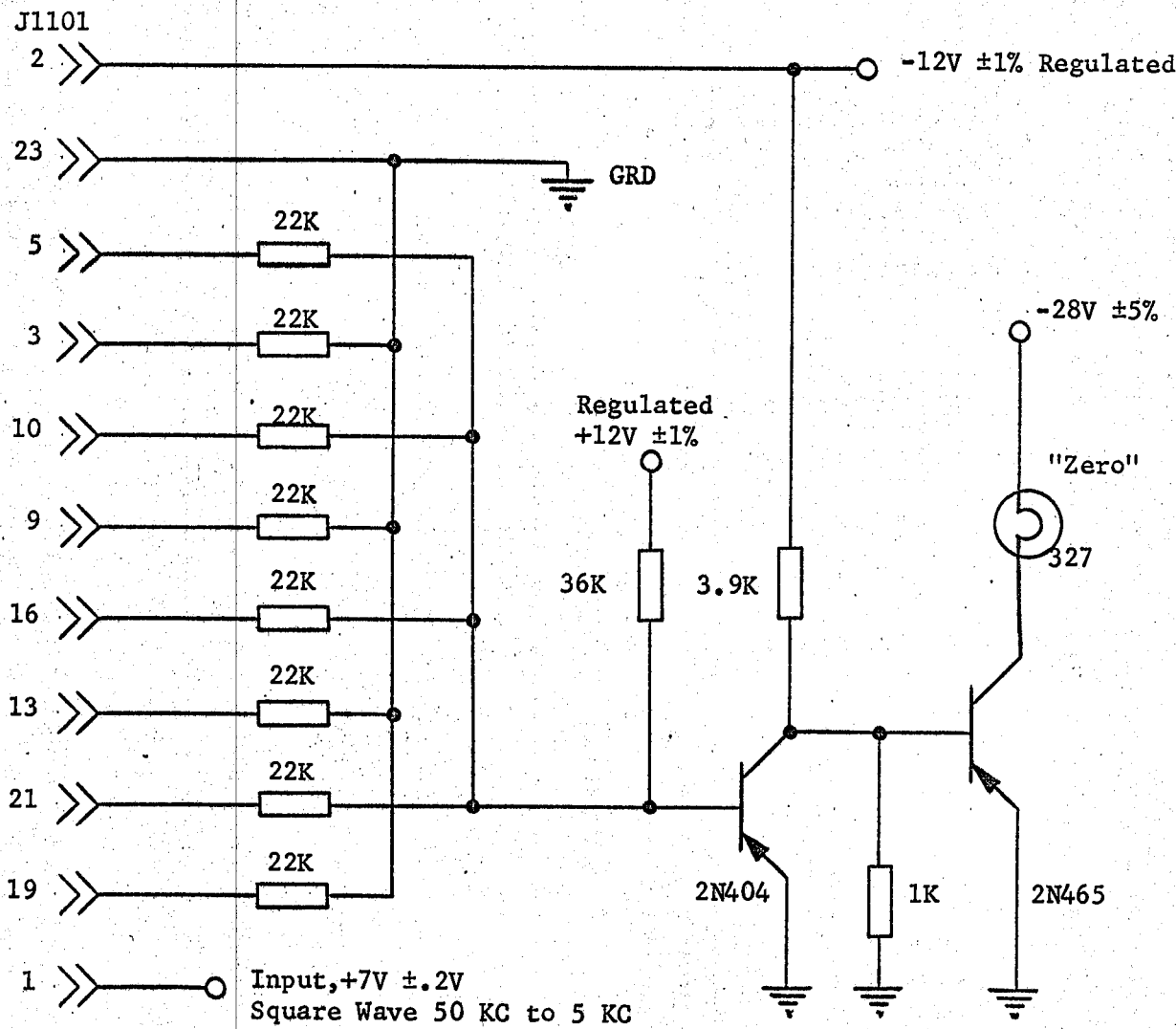


*Note: -12V must be at least 0.1 volt greater value than the +12V value -- examples; -12V and +11.9V; or -12.6V and +12.5V, or -11.6V and -11.4V.

COMPARISON CONVERTER TEST CIRCUIT

FIGURE 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
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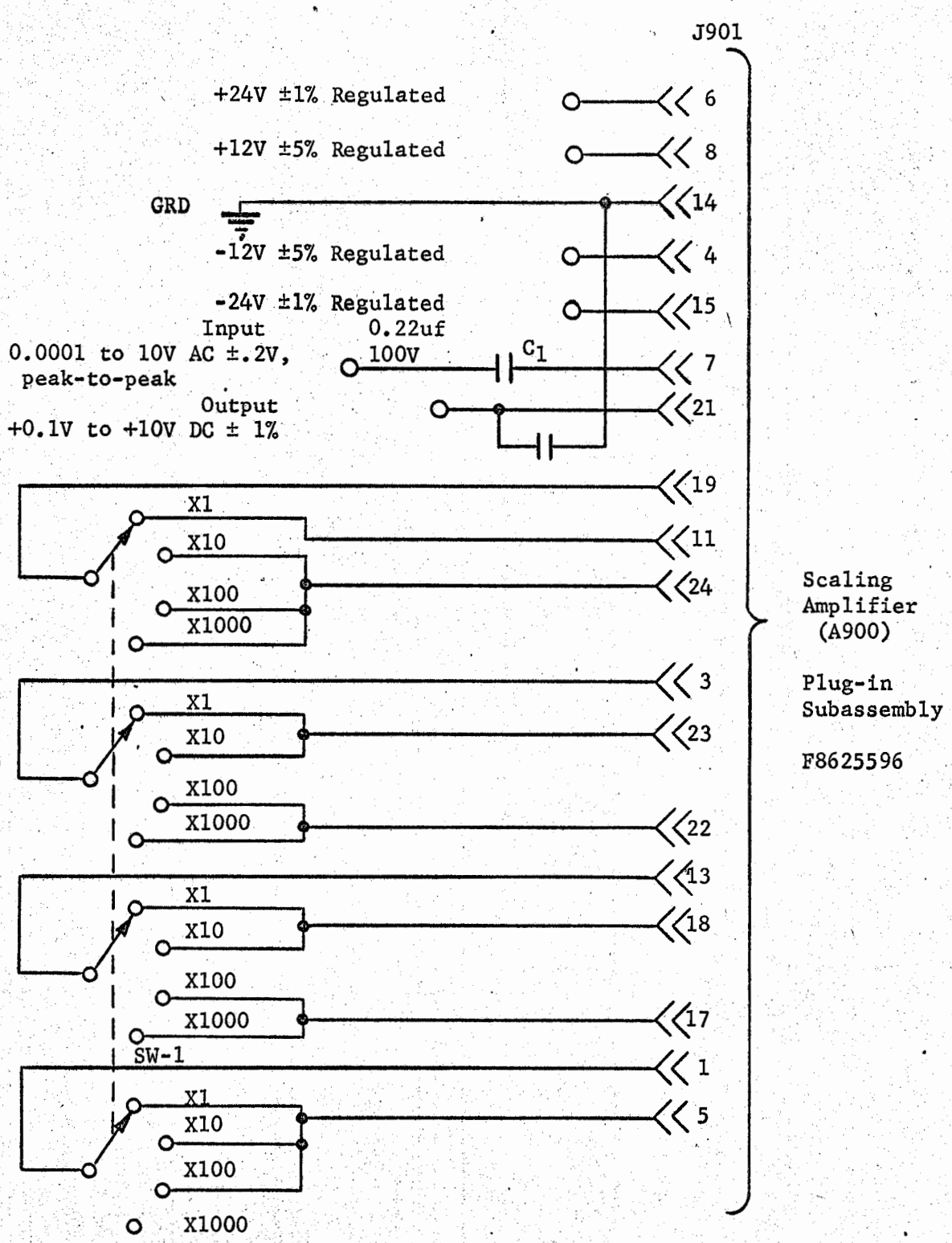


Note: All resistance values are in ohms, 1/2w, 5%.

READOUT COUNTER (A1100) PLUG-IN SUBASSEMBLY F8625341

FIGURE 3

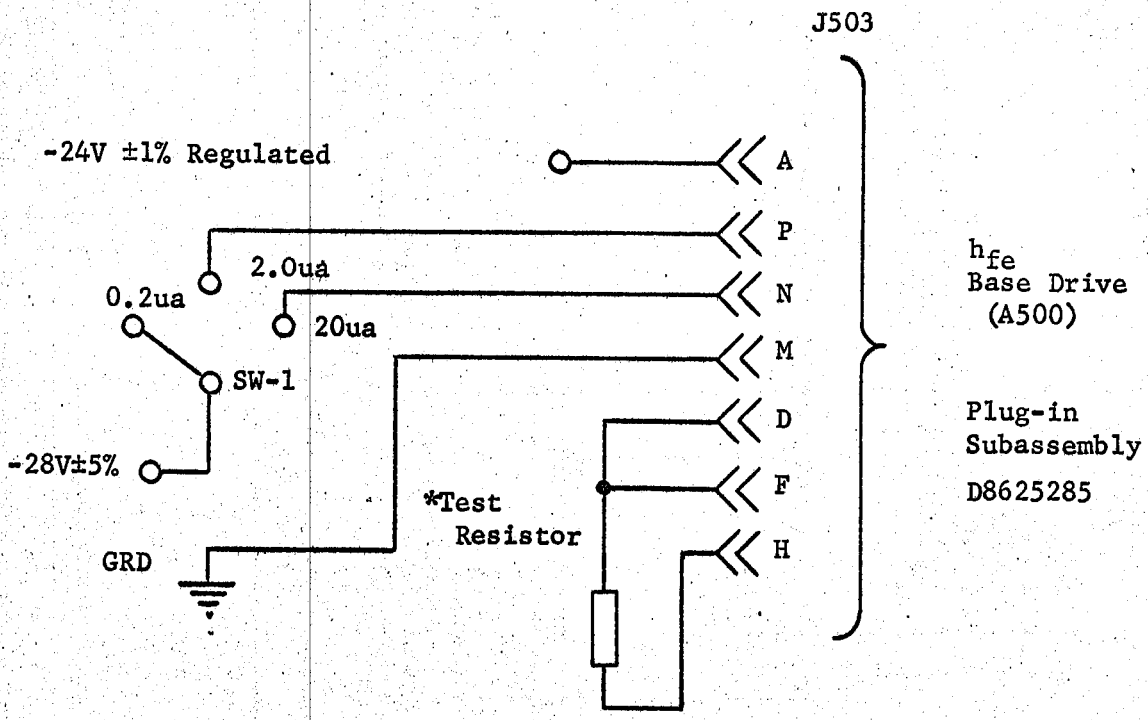
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SCALING AMPLIFIER TEST CIRCUIT

FIGURE 4

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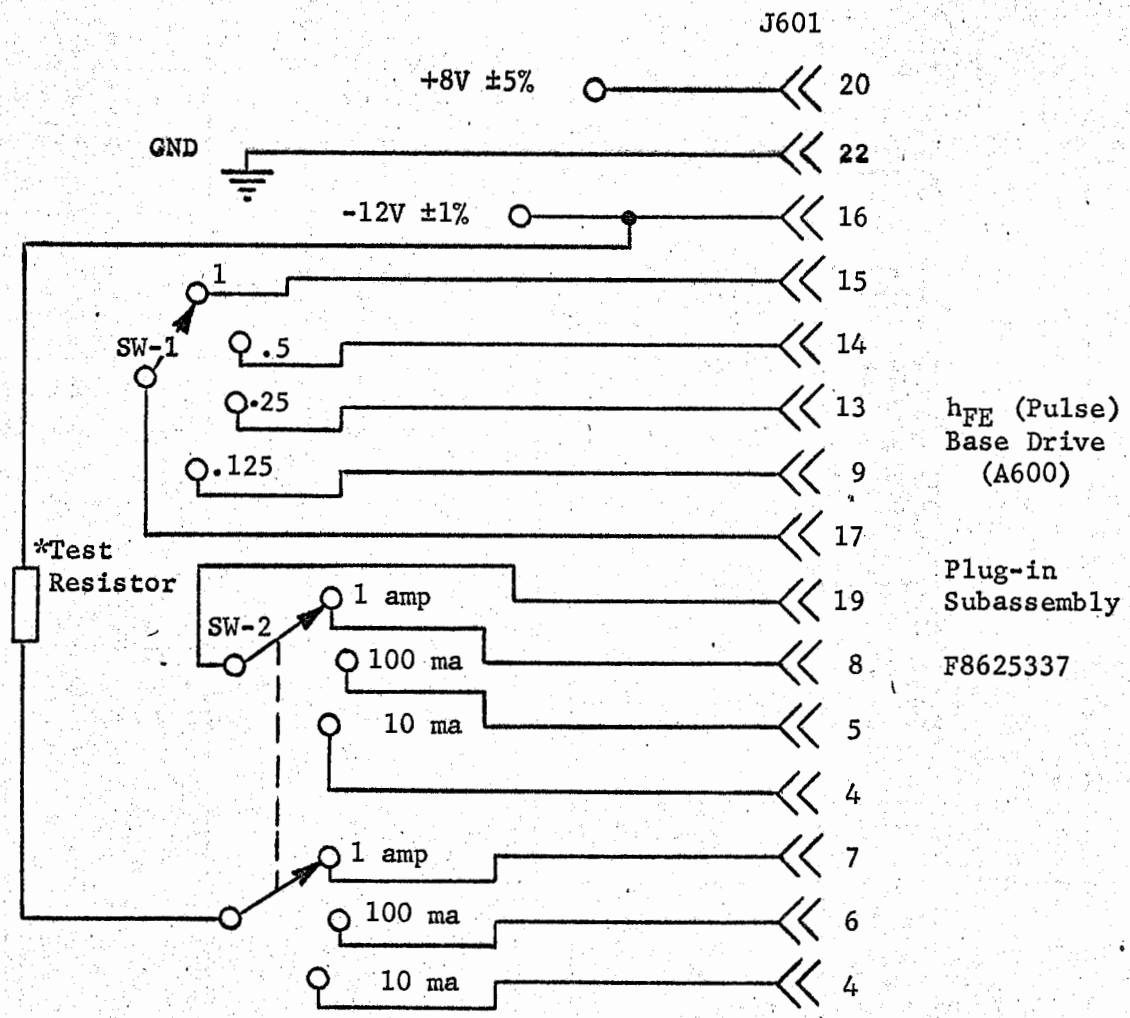


*Test Resistors are 100K, 10K, and 1K, all 1/2w, ±1%

h_{fe} BASE DRIVE TEST CIRCUIT

FIGURE 5

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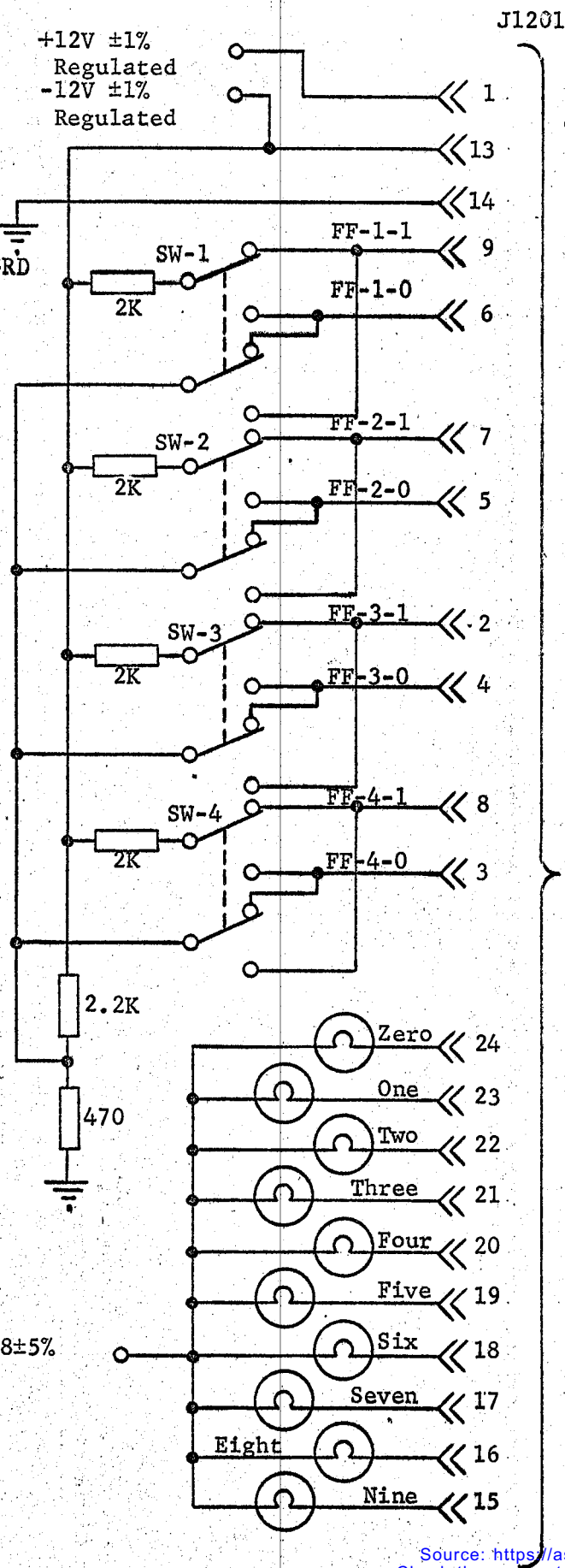


*Test Resistors will be 1 ohm ±1%, 10 ohm ±1%, or 100 ohm 1/2 w, ± 1%

h_{FE} (PULSE) BASE DRIVE TEST CIRCUIT

FIGURE 6

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Modified Binary Conversion Chart

Readout Decimal	SW #1	SW #2	SW #3	SW #4
0	0	0	0	0
1	1	0	0	0
2	0	1	0	0
3	1	1	0	0
4	0	0	1	0
5	1	0	1	0
6	0	1	1	0
7	1	1	1	1
8	0	1	1	1
9	1	1	1	1

All bulbs, Type 327

NOTES

1. SW-1 thru SW-4 shown in "Zero" position. Reverse switch for a binary "ONE".

2. All resistance values are in ohms, 1/2w, 5%.

LIGHT DRIVER (A1200) PLUG-IN SUBASSEMBLY F8625342

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FIGURE 7